Zhang, Wei

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Chinese citizen; Singapore Permanent Resident

Education

09/2013 - 06/2017 **B.S.**, *Engineering Physics*, **Tsinghua University**, Beijing, China.

09/2017 - 01/2019 Program of Interdisciplinary Courses in *Computer Science*, **University of Chinese Academy of Sciences**, Beijing, China.

09/2017 - 06/2022 **Ph.D.** in *Particle Physics and Nuclear Physics* (*major in Medical Image Analysis*), **University of Chinese Academy of Sciences**, Beijing, China. **M.S.** received in 2019.

Work Experience

2015 Research Assistant at the student training project in **Tsinghua University**, Beijing, China.

2016 - 2021 Internships at GSAFETY Technology Co., Ltd. (2016), and at the departments of neurology of Beijing Xuanwu Hospital (2019) and Beijing Tiantan Hospital (2021), Capital Medical University, Beijing, China.

2022 - present Research Fellow at Nanyang Technological University, Singapore

Representative First-Author Papers & Posters

1. **Zhang, W**., et al., Adult-to-infant unidirectional neural coupling mediates selective social learning in infants from the United Kingdom and Singapore. *Nature Communications*, 2026
2. **Zhang, W**., et al., Spatiotemporal Encoding Signatures of Covert Speech Revealed by Source-Localized EEG and fMRI. *Communications Biology*, 2026
3. **Zhang, W**., et al., Revealing the Spatiotemporal Brain Dynamics of Covert Speech compared with Overt Speech: A Simultaneous EEG-fMRI Study. *NeuroImage*, 2024
4. **Zhang, W**., et al., Investigating Sea-Level Brain Predictors for Acute Mountain Sickness: A Multi-Modal MRI Study before and after High-Altitude Exposure. *American Journal of Neuroradiology*, 2024
5. **Zhang, W**., et al., Deep Learning With 18F-Fluorodeoxyglucose-PET Gives Valid Diagnoses for the Uncertain Cases in Memory Impairment of Alzheimer's Disease. *Frontiers in Aging Neuroscience*, 2021
6. **Zhang, W**., et al., Modeling brain functional networks using graph neural networks: A review and clinical application. *ICCK Transactions on Intelligent Systematics*, 2024
7. **Zhang, W**., et al., Speaker-Listener Neural Coupling Explains Gating Effect of Gaze on Infant Statistical Learning. *OHBM Annual Meeting*, 2025

8. **Zhang, W**., et al., Multi-Site Voxel-Based Morphometry: Reliability Evaluation and a Harmonization Method. *International Conference on Medical Imaging Physics and Engineering (ICMIPE 2019) & The 9th National Annual Meeting of Medical Imaging Physics (NAMMIP)*, 2019

   https://scholar.google.com/citations?user=Kqc41x4AAAAJ

Research Interests

Multimodal neuroimaging

AI and brain-computer interfaces

Neurolinguistics

Language and social neuroscience

Reviewing & Teaching Activities

Invited reviewer for *NeuroImage*, *Imaging Neuroscience, Developmental Cognitive Neuroscience, Scientific Reports, Frontiers in Human Neuroscience, IEEE Journal of Biomedical and Health Informatics, Philosophical Transactions of the Royal Society B, Computers and Electrical Engineering, Computers, Materials & Continua, and Signal, Image and Video Processing*

2024 - 2025 Guest lecturer in *HP4021 Advanced Neuroscience Seminar,* Nanyang Technological University

2026 AI tutorial instructor in *GP8000 AI Literacy, Graduate College PGR ICC,* Nanyang Technological University

Professional Membership & Talks

2024 - 2025 Member of American Society of Neuroradiology (ASNR)

2026 Invited Young Speaker, AP-CCN 2026, Singapore

Awards and Honors

2013 Leading talent scheme, Tsinghua University, Beijing, China.

2018 - 2021 National Scholarship for Postgraduate (¥10,000 direct), Institute of High Energy Physics, Chinese Academy of Sciences

2022 Dean's Scholarship (¥13,665 direct, TOP 1 in each department, 300 out of all postgraduates in CAS), Chinese Academy of Sciences

2022 Zhu Li Yue Hua Excellent Doctoral Student Award (¥5,000 direct, TOP 1 in each department, 300 out of all postgraduate candidates in CAS), Chinese Academy of Sciences